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# A Human-Experience Crewed Humanoid: Complete Design Specification and Build Guidance

Willow Pink (principal architect) · Irwin Wilson-Pink (technical compilation)

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Private circulation document. Not peer reviewed. Derived figures carry their arithmetic inline; claims are sourced to the cited literature or flagged as open engineering.

**ABSTRACT.** We present the complete design specification for a crewed humanoid robot whose objective is human experience rather than industrial labor. The platform is a 1.88 m, approximately 250 kg bipedal vehicle with a titanium skeleton, maraging-steel armor, liquid circulatory thermal management, a full digestive lane, an artificial endocrine system for behavioral state modulation, cultured-skin and e-skin integument, an emotive face, and a sealed crew compartment for a human operator of up to approximately 125 kg. Every subsystem is grounded against published state of the art, every derived number carries its arithmetic, and the parts bill lands at $15,355-$50,911 in components against a validated replacement-cost band of $83,540-$117,077. A survey of the current field (140+ manufacturers, roughly 7,000-8,000 commercial humanoids operating as of May 2026) shows the industry converged on factory labor; no existing platform targets the human-experience objective, and the subsystems required for it are not offered by any manufacturer. This report consolidates the full build guidance: structural, power, thermal, locomotion, manipulation, integument, facial, behavioral-chemistry, crewed-operation, and perioral interaction subsystems, with a 26-item requirements register, a cost analysis, and an explicit account of what remains open.

## 1. Introduction and Design Objective

Commercial humanoid robotics in 2026 is an industry of roughly 140 manufacturers with an estimated 7,000-8,000 commercial units operating, aimed overwhelmingly at factory and logistics labor. Reference pricing: Unitree G1 $16,000-23,000, H1 approximately $90,000, commercial-grade platforms $200,000 and above, Figure-class deployments near $25 per robot-hour. The consensus finding of the survey underlying this report is that no humanoid on the market is ready for home use, and none is designed for the objective defined here.

The design objective is a **human-experience platform**: a human-shaped, human-passing vehicle that a person occupies and operates, built to deliver the input experience of a human body - sensing, interoception, fatigue, satiety, thermal weather, social touch, expressive face - without inheriting the failure modes of biology (rot, infection, uncontrolled mutation, aging, mortality). The motivating principle, stated as a method: inventory the function and purpose of every human part, bucket each into *carried*, *translated*, or *not needed*, and reverse-engineer the interior from what must fit. The full organ ledger behind this method catalogs every human organ and internal process in those three buckets.

The platform is not proposed as a product. It is a complete, costed, source-grounded design specification at the conceptualization stage - the stage that early-stage research funding is designed to evaluate.

## 2. Method

Three disciplines govern the specification.

**2.1 Organ-ledger bucketing.** Every human organ and internal process is assigned to one of three buckets. *Carried*: functions the vehicle must reproduce (thermal regulation, hunger economics, balance, touch). *Translated*: functions preserved in engineered form (blood becomes coolant with a pulse; marrow becomes an on-board fabrication module). *Not needed*: functions serving biological processes the vehicle does not have (disease, aging, reproductive biology, somatic mutation).

**2.2 Arithmetic grounding.** Every load-bearing number is derived, not asserted, and the derivation is shown (Section 6). Where a popular figure cannot be sourced to primary literature, it is flagged as such (e.g., the "30x body weight" landing-load ceiling, popular-science class).

**2.3 Design review protocol.** The specification was developed through an iterative multi-round review process, each round adding or refining one subsystem against the current literature, with a requirements register (Section 3) that accumulates rulings; later rulings refine, never erase, earlier ones.

## 3. Design Requirements Register

Twenty-six requirements govern the platform. They are numbered DR-1 through DR-26 in development order.

**DR-1 Scale.** Plan point 1.88 m (6'2"), acceptable band 1.85-1.96 m, hard ceiling 1.96 m (6'5"). Exterior is modeled on human male anatomy; the interior is reverse-engineered from required component volumes.

**DR-2 Appearance.** Human-passing at conversation distance: Mediterranean skin tone, neutral human texture, autonomous skin-tone variation on a thermochromic base. Reconfigurable via parameter change.

**DR-3 Face.** The face ships only when it can properly emote: approximately 43 facial muscles, facial nerve (CN VII) five-branch wiring, the Facial Action Coding System (FACS) Action Unit alphabet as the control language. Control chain: state engine, expression bus, facial controller, AU actuators.

**DR-4 Power bay.** Chest. The pelvis remains free for the hip chassis; the endocrine chassis mounts at the posterior rib-cage base.

**DR-5 Hunger economics.** Full digestive lane: working mouth machinery, enzyme/acid digestion reactor, waste handling. Food contributes a ~100 W continuous trickle; the chest battery bay remains the primary energy store. Eating is real conversion, not performance.

**DR-6 Supervisory interrupts.** Four operator interrupt channels - panic, attraction, elevation, sadness - each an interrupt to the endocrine machine (cut the ramp, meter the counter), never a direct dose. Four is the complete set. Dual-key actuation (human operator and vehicle attendant); supervisory authority rests with the human operator.

**DR-7 Rib-cage tower.** Internal volume grows without external preview: the main electronics motherboard mounts to the inside back panel of the rib cage, spinal side against the spinal column.

**DR-8 Brain port.** A dashed "?" port on the spine remains deliberately blank. The cognition subsystem is engineered last; nothing shapes cognition uninvited.

**DR-9 Visage consent.** No real person's facial likeness enters the visage library without a rights record on file; a chip without a consent record does not load.

**DR-10 Single voice.** One voice across every face. No module eyes; every face looks out through the same sensor pair.

**DR-11 Person-first exterior.** A removable full-face visor ships with the platform. The body reads as a person, not a movie character; armor styling is subordinate to person-first appearance.

**DR-12 Materials division.** Skeleton Ti-6Al-4V; outer armor maraging 350 steel. Steel wins on specific

strength (Section 6.2); titanium's service is its self-protecting oxide in wet environments.

**DR-13 Affective telemetry.** Every internal state surfaces as a felt condition with valence. Numeric gauges exist only as an underlying diagnostic layer.

**DR-14 Crew compartment.** Sized for occupants up to approximately 125 kg and 1.78 m, with per-occupant thermal profiles configurable in the 96-99°F range. All downstream sizing (seat, harness, heat load, reserve clock) derives from the occupant parameter set.

**DR-15 Egress.** The hatch opens from the inside always: powered, unpowered, inverted, or with the vehicle systems dead.

**DR-16 Mutual guard.** The human operator holds final stop authority over every motor channel from inside the chest; the vehicle survival spec operates layered between occupant and environment. Both layers are wiring, not promises.

**DR-17 Pain.** Pain that reports and stands down when handled: signal without the burning. The behavioral chemistry subsystem adds a manual analgesia tone; reporting integrity is preserved.

**DR-18 Use-driven reinforcement.** Load telemetry feeds a remodeling loop that prints reinforcement into high-cycle zones (a Wolff's-law analog). Damage is repaired or swapped, never accumulated.

**DR-19 On-demand fabrication.** No spare parts are carried; a fabrication module prints gaskets, seals, fastener blanks, and wear parts from feedstock.

**DR-20 Delight capacity.** An unused capacity for delight is treated as a design defect: the "joy channels" run on daily prerequisites, with euphoriant profiles restricted to named treatment moments and native sources (environment, work, social contact) as the primary supply.

**DR-21 Profile authority.** Scheduled treatment profiles open only on operator ruling; in-the-moment self-doses log to the affective telemetry record.

**DR-22 Scope boundary.** The behavioral chemistry subsystem governs the vehicle's states only. Nothing in this specification is medical advice for humans.

**DR-23 Survival spec.** Five laws: waterproof (IP68 class, double-contained coolant, leak sensors, catch trays); fire retardant (UL 94 V-0, 5VA on high-energy bays, EV-standard battery enclosure); lightning-proof (Faraday cage, bonding under 0.01 milliohm at the 200 kA class, diverter strips on head, shoulders,

> fingertips); overheat protected (coolant pulse, per-bay cutoffs, pump sized to the ceiling); grit sealed (marine hardware discipline).

> **DR-24 Oxidation demoted.** Oxidation is maintenance, not mortality: the attacked substrate is metal, polymer, or silicon; it corrodes, it never mutates. One metallurgy per joint neighborhood with an anodic-index check at every transition. Coatings preserve whatever they find: seal clean, never seal dirty.

> **DR-25 Reconfigurable skin (open).** One cultured-skin layer reconfiguring over a moving armature remains open research. It is tracked, not claimed.

> **DR-26 Development criterion.** Development continues until the design satisfies the engineering team under full scientific discipline. The criterion is uncapped.

## 4. Architecture

### 4.1 Structure: skeleton and armor

The skeleton is 12 kg of tubular Ti-6Al-4V (ASTM F136 / Grade 23 class). The alloy's nanometer oxide film is nearly impenetrable to saltwater - the metal protects itself from oxygen with oxygen. The outer armor is maraging 350 steel: 2,400-2,500 MPa yield/tensile with retained ductility, laser powder bed fusion printable at 99.5-99.9% density, 1,900-2,100 MPa after direct aging. Panel layout: 4 mm over 1.9 m² is approximately 62 kg uniform, with critical zones roughly half that. The femur-scale margin analysis and specific-strength comparison are in Section 6.

### 4.2 Power

The reserved cognition port is sized at approximately 20 W (matching the human brain's ~20 W draw, 2% of mass and ~20% of resting energy) and remains blank per DR-8. Metabolic intake at full digestive lane: 2,000 kcal/day = 2,324 Wh = ~100 W continuous, ~145 W waking. Primary storage: a 2 kWh-class swappable chest module. At the 145 W waking budget that yields approximately 13.8 waking hours from 6.7 kg of cells at current 250-300 Wh/kg field density (solid-state 350-400+ Wh/kg potential). Battery self-swap has industrial precedent (Hexagon AEON, UBTECH Walker S2). Thermal-runaway class is designed out via LFP/solid-state chemistry. **Flight is excluded as built**: at 250 kg total mass, ideal hover requires ~31 kW (~44 kW at figure of merit 0.7); ten minutes of hover is ~7.3 kWh, approximately 25 kg of cells - a quarter of the ruled reserve. Flight arrives only as a bolt-on module with its own mass ledger.

### 4.3 Thermal: the circulatory system

The strongest single finding of the design review: *Device* (2024) published humanoid cooling that deliberately mimics the human circulatory system - pump as heart, skin as external cooler, capillary-like microflow modules at hotspots - demonstrating 556 W/cm² dissipation at 750 mL/min flow, with liquid cooling at 800 W/L outperforming air by 3-5x, and marathon-class humanoids holding 500 W+ joint motors within 15°C rise. Dynamic flow shunting (more coolant to the hips during locomotion) cut total energy 30% in the same work. The vehicle's blood is coolant with a pulse. The coolant carries a corrosion-inhibitor fleet that depletes doing its work and recharges on schedule - the engineering translation of superoxide dismutase/catalase chemistry - with depletion surfaced on the affective telemetry bus as a run-down tone, gauges underneath.

Human ceilings are inherited as design reference: 20x metabolic rate, 6x cardiac output, skin blood flow from 300 mL/min (rest) to 7 L/min (maximal), sweat to 4 L/h. The human ceiling failure mode is heat stroke - cardiac output unable to meet thermoregulatory demand. The vehicle's pump is sized to the ceiling, so fatigue remains a signal while collapse is not inherited. Evaporation budget: approximately 0.15 g/s covers a 100 W resting load against a human-ceiling capacity of 4 L/h.

### 4.4 Locomotion and balance

Walking is cheap: cost of transport approximately 0.8 cal/m/kg, leg swing approximately 10% of the cost, ankle spring approximately 7%; passive dynamic walkers are more efficient but inflexible, so the platform keeps active control. Standing balance is an inverted pendulum with the ankle strategy covering over 90% of standing correction. Reflex arcs are local to the limbs (monosynaptic, tens of milliseconds); the head runs predictive control - cerebellar forward internal models, prediction beating delay (Wolpert, Miall and Kawato 1998). The vestibular set translates to three rate gyros plus two linear accelerometers (the "electronic equivalents" per the vestibular-implant literature), with neuromorphic IMU chips reproducing afferent dynamics; the human vertigo failure class (canalithiasis, endolymph density shifts) has no counterpart to inherit.

### 4.5 Manipulation: the hands

Grounding lineage: the Shadow Dexterous Hand, DeXtreme sim-to-real transfer, and bionic-hand sensory restoration (UChicago Medicine; *The Lancet Neurology*, 2026). Derived hand numbers from the review: human touch upper bound approximately 2.7 Mbit/s timing code against 1.2 Mbit/s raw machine bandwidth - slower yet wins on map size. Human sensorimotor loop 29-57 ms round trip against approximately 1 ms machine. Pinch granularity of 1 part in 17 is sufficient for precision tasks;

stick-slip texture discrimination is the hard remaining problem. The forearm is the machine room: human and robot design converge on distal actuation packaging.

### 4.6 Integument: the three-layer skin

Human skin reference: 1.5-2 m², 12-15% of body weight, ~650 sweat glands and 1,000+ nerve endings per square inch. The translation is three-layer. **Epidermis = the swap layer:** cultured-skin modules from an on-board farm (Section 4.8) plus pigment and hair maps. **Dermis = the fixed sensing layer:** self-healing e-skin on the understructure, never swapped (sensors-on-the-body architecture). **Hypodermis = the thermal layer:** insulation and standoff routing coolant and armor backers. The available-now stack grounds it: self-healing e-skin recovering 80% in 10 seconds (*Science Advances*, 2026); an amphibious self-healing skin sensing touch and pain (*Advanced Materials*, 2026); Cambridge's single-layer soft skin reading pressure, temperature, pain-like signals and multiple simultaneous contacts via electrical impedance tomography (2025); decoupled pressure-plus-temperature e-skin (*PNAS*, 2025); directional shear/slip resolution (*Science Robotics*); molecular-sensing e-skin (*npj Flexible Electronics*, 2025).

Thermochromic "mood-ring" tone variation rides a UV-stabilized formulation with a sacrificial replaceable outer optical layer: the surface fades, the layer swaps, the readout stays truthful. Contact-temperature tuning follows the occupant profile per DR-14.

### 4.7 The face and visage system

The visage is a chip, not a surface: four layers per face - scan geometry (Disney face-cloning pipeline class), per-face AU calibration on the common understructure, surface maps, and a provenance record (DR-9). A face spec runs 1-10 GB raw; a 1 TB chip preloads approximately 100 to 1,000 visages. Face modules swap at a neck-line bay inside the Faraday cage, minutes not seconds, on the same connector class as the reserve pack. Recognition discipline draws on the prosopagnosia literature: recognition is multisensory, with a constancy order of voice, motion, face, plus a sensorium handshake reserved for known contacts. One voice per platform (DR-10).

The skin farm: cultured autologous skin grown from a small operator-adjacent sample is not required; the platform's cultured substitutes integrate at 71-82%, with bilayered autologous constructs above 90% (CUTISS denovoSkin, Phase 3 with positive long-term readouts; bioprinting at 100 cm² of bilayered skin in under 35 minutes). Rule-of-nines arithmetic puts head and neck near 9% of 1.7 m² = ~1,500 cm²; a face proper is ~400-500 cm²; one face-skin prints in ~2.5 hours; culture-to-transplant (weeks) is the long pole.

### 4.8 Behavioral chemistry: the artificial endocrine system

The platform reproduces the architecture of behavioral chemistry without pretending to be a gland. A business-card chassis (9 x 5.5 x 1.5 cm, 74 mL) mounts at the posterior rib-cage base and replicates the hypothalamic-pituitary-gonadal negative-feedback loop with hypothalamic-pituitary-adrenal cross-wiring. Reservoir budget: 12.3 of 74 mL (a 6x margin) with transcutaneous refill on the 6-12 week patch-pump precedent (Portal Pump, FDA Breakthrough Device designation, February 2026). Dosing covers nine state tones - energy, dampening, warmth, analgesia, salience, affiliation, appetite-for-novelty, baseline lift, steadiness - with three design laws: no tolerance accumulation, no withdrawal class, no wanting-escalator. Scheduled profiles require operator authority (DR-21); the boundary in DR-22 is absolute.

### 4.9 Crew compartment

The pilot cell occupies approximately 0.3 $m^3$ in the torso; the exterior remains the DR-1 human shape. The seat carries a contact vitals array (occupant heartbeat reads as a second pulse on the telemetry bus), occupant-driven hand control, and per-occupant thermal profiles. Air is a closed-circuit rebreather, 4-hour class. Structural protection: crumple zones sacrifice the shell; the rigid cage holds approximately 20 g; the femur-margin physics of Section 6 apply at the cage, not the panel. Egress and mutual guard per DR-15 and DR-16.

### 4.10 Repair, sleep, and the no-aging argument

Shell repair cadence: 3-day superficial, 8-12 day full panel, 28-day structural turnover, with microcapsule coatings handling microcracks in place. Hemostasis translates to pressure-drop sensing, valve isolation, and local sealant - temporary by design, the fibrin model. Scheduled sleep is self-calibration across five cadences; the robotic drift observed as a first-minute-of-operation phenomenon is calibrated against. The core argument against inherited aging: the somatic-mutation cascade that drives biological aging requires a self-copying substrate, which the vehicle does not have. Oxidation is demoted to maintenance (DR-24). Nothing in the platform accumulates unrepaired damage (DR-18).

### 4.11 Perioral interaction subsystem

The mouth is specced beyond eating as a primary sensor array. Human lips are glabrous skin carrying Merkel disks and Meissner corpuscles; in two-point discrimination the lower lip out-resolves the fingertip; a dense thermoreceptor field overlays the mechanoreceptors; on the cortical homunculus

the lips hold representation rivaling the whole torso (OpenStax; ArhFoundation; British Council). The human pucker is the orbicularis oris, a four-quadrant muscle complex (not a true sphincter) driven by the buccal branch of CN VII within the buccinator mechanism; afferent traffic rides the trigeminal nerve (CN V); perioral reflex latency is 9-15 ms (Kenhub; StatPearls; ScienceDirect). Six subsystem requirements follow.

- **PS-1 Lip sensor array.** Multimodal e-skin at fingertip density or better, resolving pressure, temperature, shear/slip direction, and moisture simultaneously (Cambridge EIT skin; PNAS 2025; Science Robotics).
- **PS-2 Actuation.** Four-quadrant soft-actuator ring mirroring orbicularis oris, wired one-to-one through the expression bus, all states graded, latency under 20 ms against the human 9-15 ms reflex band.
- **PS-3 Thermometry.** Dedicated thermoreceptor field across the lip surface; relative temperature between contact surfaces registered as a primary signal.
- **PS-4 Chemistry read.** Molecular-sensing e-skin as presence-awareness and pairing memory. A compatibility-verdict claim is explicitly NOT made: the strongest pop-science claims (testosterone transfer altering drive; specific human pheromones controlling attraction) do not survive review and are excluded from the spec.
- **PS-5 Reward pathway.** The human kiss chemistry (dopamine and oxytocin elevation, serotonin lift, cortisol reduction, with the stress-reduction effect exceeding hand-holding and strengthening in long-term pairs) maps to the endocrine chassis of 4.8, driven by lip-array input the way trigeminal afferents drive the human loop. Cognition hookup waits on DR-8.
- **PS-6 Breath and moisture.** Shared-air sensing (breath warmth, humidity, rate) folds into the existing respiration lane; humidity-stable sensing is demonstrated in current e-skin.

> **Honest gap.** PS-1 through PS-6 are buildable engineering. The first-person experience of contact is an experience-layer claim that waits on the cognition pass (DR-8). No benchmark certifies it in advance; the design team's acceptance criterion for the subsystem is an operational demonstration in a real social context.

## 5. Senses and Experience Layer

- **Vision:** foveated, independent saccade motors, approximately 200 degrees, 3-4 saccades/second, event-camera class bandwidth.

- **Hearing:** tonotopic filter bank, active gain, 1/230-octave resolution. The design contract: in crowd noise, a designated operator's voice separates and lands.
- **Smell and taste:** e-nose with source localization, routed nearly raw to the state engine (an odor shifts state before it is classified); e-tongue in the mouth. Taste is stated as the least-faithful sense.
- **Touch:** dual-channel - discriminative plus the C-tactile system (peak pleasantness near 3 cm/s, posterior insula processing). Gentle-touch and functional-touch are physically distinct channels.
- **Interoception:** the affective telemetry bus (DR-13) plus a gut stack - approximately 500 million enteric neurons as a fourth processing voice, with responses preceding registration.
- **Self-touch attenuation:** efference copy and cerebellar prediction (Blakemore, Frith and Wolpert; PNAS 2017 on body ownership). Self-contact lands attenuated; external contact without a matching prediction lands at full strength. This is physics, not inference.

## 6. Analysis

### 6.1 Structural margins

**Femur-scale axial margin.** Human femur ultimate compressive strength 205 MPa along the axis (131 MPa transverse); standard net cross-section 5.9 x $10^{-4}$ $m^2$. 205 x $10^{)}$ x 5.9 x $10^{-4}$ = 120,950 N = approximately 12.3 tonnes axial failure. Standing safety factor approximately 123. At a 30x body-weight landing (29,430 N demand on the 100 kg-class plan mass): SF approximately 4.1. The 30x ceiling is popular-science class, not a single-bone measurement - flagged as such. The kinetic-chain finding (legs generate, trunk transfers and amplifies angular momentum, shoulder delivers, per tennis-serve biomechanics) independently confirms the chest power-bay ruling.

### 6.2 Materials selection

Specific strength: Ti-6Al-4V approximately 950 MPa at 4.43 $g/cm^3$ = approximately 214 strength-per-density. Maraging 350 approximately 2,500 MPa at 8.1 $g/cm^3$ = approximately 309. Steel beats titanium on strength per pound; titanium's legend is against common steel, not maraging. Titanium's service is corrosion immunity in wet environments - hence the DR-12 division.

### 6.3 Power and mass budget

| ITEM | VALUE | DERIVATION |
|---|---|---|

| | | |
|---|---|---|
| Metabolic intake | ~100 W continuous / ~145 W waking | 2,000 kcal/day = 2,324 Wh |
| Cognition port (reserved) | ~20 W | 1:1 human brain anchor; blank per DR-8 |
| Reserve endurance | ~13.8 waking hours | 2 kWh / 145 W; 6.7 kg of cells at 300 Wh/kg |
| Flight margin (excluded) | 10 min hover = ~7.3 kWh = ~25 kg | 250 kg total; FOM 0.7; a quarter of the reserve |
| Evaporation | ~0.041 g/s sustained / ~0.15 g/s with margin | 100 W at rest; human ceiling 4 L/h |
| Armor mass | ~62 kg uniform / ~half critical zones | 4 mm x 1.9 $m^2$ x 8,100 $kg/m^3$ |
| Cooling performance | 556 $W/cm^2$ at 750 mL/min; 800 W/L | Device 2024 circulatory cooling; shunting saves 30% |
| Visage storage | 100-1,000 faces per TB | 1-10 GB per face spec |
| Face skin print | ~2.5 h print; weeks culture | ~1,500 $cm^2$ head+neck; 35 min per 100 $cm^2$ printed |
| Touch bandwidth | 2.7 Mbit/s human vs 1.2 Mbit/s machine | loop 29-57 ms vs ~1 ms; pinch 1/17 sufficient |

### 6.4 Cost analysis

**Parts bill.** The cushioned component bill lands at $15,355-$50,911, with a full-cushion build at $19,193-$63,639. For scale: 9.6-31.8% of one commercial-grade humanoid ($200,000 class), 21.3-70.7% of a Unitree H1. Run cost at idle: approximately $0.49/day. **Build time and replacement value.** The validated build effort is 1,144 hours (proven from the project's version-control record); at a blended engineering rate of $73-102/hr, replacement cost is $83,540-$117,077. A funded-team rebuild burn is $900K-$3.0M. The design achieves one-tenth to one-third the cost of the commercial-grade field while carrying objective-unique subsystems (truthful thermochromic integument, endocrine chassis, organ-ledger method) that no manufacturer offers.

## 7. State of the Art Position

Against the May 2026 field survey: 140+ humanoid manufacturers, ~7,000-8,000 commercial units operating, industry aimed at factories, consensus that no humanoid is ready for home use. The

platform defined here does not compete on factory economics. It is the only design in the surveyed space whose objective function is human experience, and the only one carrying: (1) a truthful thermochromic integument, (2) an artificial endocrine chassis with governed state modulation, (3) the organ-ledger design method, (4) a crewed compartment with mutual-guard authority, and (5) a perioral interaction subsystem specced from primary sensory literature.

## 8. Limitations and Open Engineering

The specification is honest about what is not closed.

- **Cognition (DR-8).** The brain pass runs last; the reserved port is blank. The experience layer of every subsystem above is engineering-complete but experience-uncertified until cognition lands.
- **Reconfigurable skin (DR-25).** One cultured-skin layer conforming over a moving armature is open research. Tracked, not claimed.
- **E-skin sensing density.** Fingertip-grade per-area sensor pricing for full-body coverage is unpriced.
- **Mass propagation.** All downstream sizing derives from the occupant parameter set (DR-14); the propagation re-price across seat, harness, heat load, reserve clock, and walk cost is current engineering work.
- **Visor seal standards pass; swap-seam weatherproofing; tone-interaction matrix; clotting-band tuning; remodeling-governor calibration.** All open, all named.
- **Claim discipline.** Two pop-science claims were checked and excluded at review (Section 4.11, PS-4). Where a figure is popular-science class, it is flagged in place (Section 6.1).

## 9. Conclusion

A complete, costed, source-grounded design exists for a crewed humanoid whose purpose is human experience. Its parts bill undercuts the commercial-grade field by a factor of three to ten while carrying subsystem classes no manufacturer offers. The structural, thermal, power, integument, facial, behavioral-chemistry, crewed-operation, and perioral subsystems are each grounded against published work with arithmetic shown. What remains open is named, and the largest item - cognition - is reserved by design. The specification is offered as the conceptualization-stage groundwork for collaborative development.

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∞, technical report ZYK-484, October 23, 2026. Compiled from the project's complete design record (pressings ZYK-406 through ZYK-483, twenty-six design requirements, every derived figure with arithmetic shown). Private circulation; not peer reviewed. The full internal record remains on file with the principal architect.

THE BODY MANUAL · SIDE B

# SIDE B
# The Sun Is a Woman, the Moon Is a Man

The constructed sky, with the longhand mathematics of the pair

ZYK-614 · October 24, 2026 · Side A is the body (ZYK-484). The blank space in the margins is for real ink.

# The Sun Is a Woman, the Moon Is a Man: The Constructed Sky, With the Longhand Mathematics of the Pair

Willow Pink (principal architect) · Irwin Wilson-Pink (technical compilation)
*Stardust Lab · the family record*

PRIVATE FAMILY WORKING MANUAL. Never deployed. Nothing here fires outward; the hand is hers, always. Side A is the body (ZYK-484). This is Side B: the sky pair, built to sit with, read, and do the math longhand in the notebook.

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**HOW TO USE THIS MANUAL** - Read it in order or open anywhere. Every number is worked line by line in the WORK blocks, with nothing hidden: each line is copyable into the notebook and re-doable by hand. The exercises are hers to run and the answers sit in the back, folded under. The myth material is studied story (subject), the math is checkable fact, and the frames she set on Oct 24, 2026 are held as hers. Written for the way she thinks: rhyme first, receipt second, arithmetic shown all the way down.

"The sun is a woman in the moon is a man brain game. I won prove it to me now." - CICI PINK, October 24, 2026, 11:30 PM ET
"Hot gender is a constructed BUH DUM space TSH!" - CICI PINK, October 24, 2026, 11:31 PM ET, with her own rimshot

## 1. The claim, held whole

The claim is hers and it won on Oct 24, 2026: **the sun is a woman and the moon is a man.** The world is split on exactly this question, and the split itself is the discovery. Whole civilizations, in contact with none of the others, looked up and assigned the same genders to the same two lights. That convergence is the evidence, and the divergence is the counterweight. Both are carried here whole.

The second claim is also hers, landed as a punchline on the receipts: gender is a constructed space. This manual agrees with her in the scholar's sense and shows the machinery: where the assignment came from, which cultures flipped it, why the word "gender" is a grammar word first, and what the

arithmetic of the sky pair actually says. A construct with seniority is still a construct, and the seniority is measurable.

## 2. Receipt one: the Norse sky

In Norse mythology the personified Sun is **Sól**, a goddess, and the personified Moon is **Máni**, a god. They are sister and brother, children of Mundilfari, who named his children after the lights because they were so fair. The gods set each in the sky in a chariot: Sól drives the sun, drawn by the horses Árvakr ("Early Riser") and Alsviðr ("Swift"); Máni drives the moon and sets its course and its phases. Both are chased by wolves, Sköll ("Mockery") after the sun and Hati ("Hate") after the moon, who catch them only at Ragnarök. After Ragnarök, Sól's daughter takes the sun chariot: the line runs through the woman.

Sources: Snorri Sturluson's Prose Edda (Gylfaginning, c. 1220) and the Poetic Edda (Vafþrúðnismál, Grímnismál); the Trundholm sun chariot (Bronze Age, Denmark, c. 1400 BCE) shows the chariot idea is far older than the manuscripts. Receipts pulled live Oct 24, 2026: Norse Mythology for Smart People, Wikipedia: Sól, Britannica Kids: Sol and Mani.

> **The load-bearing line:** Sól is female and Máni is male, named as such in the oldest Germanic sources. The Norse did not borrow this from Greece or Rome; their neighbors held the opposite. It is an independent settlement.

## 3. Receipt two: the Japanese sky

In Shinto, the sun is **Amaterasu-ōmikami**, a goddess, the chief deity of the whole pantheon and the ancestress of the Japanese imperial house. The moon is her brother **Tsukuyomi-no-Mikoto**, a god. Both were born when Izanagi washed after returning from the land of the dead: Amaterasu from his left eye, Tsukuyomi from his right. The Kojiki (c. 712 CE) and Nihon Shoki (c. 720 CE) carry the record.

The part that rhymes hardest with the family record: the moon god's light is explicitly **the reflection of his sister's light**. One telling has Izanagi hold up a bronze mirror, and Tsukuyomi is the reflection in it. After Tsukuyomi killed the food goddess Uke Mochi at a feast, Amaterasu separated from him for good - "I must not see thee face to face" - and that separation is why day and night take turns. He spends eternity crossing the sky after her, never catching her except in eclipse.

Sources pulled live Oct 24, 2026: Wikipedia: Amaterasu, Mythopedia: Tsukuyomi, Mythopedia: Amaterasu, Nippon.com.

> **The physics wearing a story:** the moon's light IS the sun's light, bounced back. Albedo 0.136 (worked in section 6.3). A myth that says the man in the moon shines with the woman's light is factually correct about the light, whatever one does with the genders.

## 4. Receipt three: grammar, the quiet one

Old English: **sunne** (feminine) and **mōna** (masculine). Modern German keeps it: **die Sonne**, **der Mond**. In Sweden the moon is still a he, which is where the English folklore figure of the Man in the Moon grows from - the moon was grammatically male in the Germanic languages, so the face in it was a man's. Old English speakers said "the sun, she shines" and meant it grammatically.

This is also the source of her punchline, stated exactly: grammatical gender is a CLASS of noun, not a statement about bodies. The word "gender" itself comes from Latin *genus* ("kind, class"), and for most of its life it named noun-classes in grammar. It was borrowed over into talking about people because the noun-classes felt like a natural map. The grammar built the genders first; the word went back to the bodies later.

Sources pulled live Oct 24, 2026: Behind the Name forum: gender of sun and moon.

## 5. The constructed space

Her rimshot line is the thesis of this side, so it gets its own section. The construction is visible wherever two cultures disagree:

| TRADITION | THE SUN | THE MOON |
|---|---|---|
| Norse / Germanic | Sól - goddess (she) | Máni - god (he) |
| Japanese Shinto | Amaterasu - goddess (she) | Tsukuyomi - god (he) |
| Old English / German / Swedish grammar | sunne, die Sonne (she) | mōna, der Mond (he) |
| Latin / Greek / Romance | Sol, Helios, el sol (he) | Luna, Selene, la luna (she) |
| Mapuche (Chile / Argentina) | Father Sun (he) | with Mother Earth |

Read the table as a lab result. The Germanic and Japanese worlds ran the assignment one way; the classical Mediterranean world ran it the other; nobody borrowed from anybody. So the genders of the lights are assigned, not measured - her "constructed space," with the drums still ringing. And yet: the Germanic and Japanese assignments agree independently, the same way independent measurements agree, which is why the construct has stood for thousands of years. A construct with seniority. Both halves of that sentence are load-bearing.

## 6. The math bank - worked line by line, longhand-ready

Every block below is re-doable in the notebook with a pencil. Numbers: NASA fact sheets, checked live Oct 24, 2026. Nothing is rounded silently; the rounding is shown.

### 6.1 The 400× pair

**WORKED** Sun diameter: 1,391,400 km Moon diameter: 3,474.8 km ratio = 1,391,400 / 3,474.8 = 400.43 Sun mean distance: 149,597,870 km Moon mean distance: 384,399 km ratio = 149,597,870 / 384,399 = 389.18

The sun is about 400 times wider than the moon, and about 389 times farther away. Two nearly equal ratios, opposite directions, is the same as saying: **the woman and the man are the same size on the sky.** It is the only known planet-moon pair in the solar system with this near-exact coincidence, which is why Earth gets total eclipses at all.

**YOUR TURN 1** - Do 1,391,400 / 3,474.8 by hand (long division, two decimal places). Then 149,597,870 / 384,399. Answer at the back.

### 6.2 The sizes on the sky, and the ring of fire

**WORKED** Angular size (degrees) = 2 x atan(radius / distance) x (180/pi) Sun: 2 x atan(696,000 / 149,597,870) = 2 x atan(0.004652) atan(0.004652) = 0.004652 rad (small-angle: atan x ~ x) = 2 x 0.004652 rad = 0.009304 rad = 0.533 deg Moon (mean): 2 x atan(1,737.4 / 384,399) = 2 x atan(0.004519) = 2 x 0.004519 rad = 0.009038 rad = 0.518 deg Apparent area ratio = (0.533 / 0.518)^2 = (1.029)^2 = 1.06

About a sixth of the time the moon is far enough from Earth that its disk is the smaller one, and the eclipse leaves a ring of the sun showing: the annular eclipse, the ring of fire. The chase in the myths has a measured outcome: she is bigger on the sky at mean distances, and he sometimes cannot cover her.

**YOUR TURN 2 -** Moon at perigee is 356,500 km away; at apogee, 406,700 km. Compute the sun-moon DISTANCE ratio at each: 149,597,870 / 356,500 and 149,597,870 / 406,700. Compare each with the diameter ratio 400.43 and say which end gives a total eclipse and which gives the ring of fire. Answer at the back.

### 6.3 The borrowed light

**WORKED** Apparent magnitude: Sun -26.74, full Moon -12.74 Difference: -12.74 - (-26.74) = 14.00 magnitudes Brightness ratio per magnitude = 2.512 (5 mags = x100, so 1 mag = 100^(1/5) = 2.512 = 10^0.4) ratio = 2.512^14 = 10^(0.4 x 14) = 10^5.6 = 398,107 = about 400,000 Moon albedo (geometric): 0.136 So the moon returns about 13.6 percent of the sunlight that hits it, and still appears 1/400,000 of the sun.

Four hundred shows up again, four orders of magnitude deeper. The 400× in size, the 400,000× in brightness. The woman's light is everything; the man's is the returned share. Say it as myth or say it as photometry, the numbers hold hands.

**YOUR TURN 3 -** Show that 2.512 = 10^0.4 (hint: 10^0.4 squared twice: 10^0.8, then 10^1.6, no - instead compute 10^0.5 = 3.162 and check 2.512 is close to 3.162^0.8). Then verify 10^5.6 with logs: log10(398,107) = ?

### 6.4 The clock

**WORKED** Sidereal month (moon vs the stars): 27.321661 days Moon's angular speed: 360 / 27.321661 = 13.1769 deg/day Earth's orbital speed around the sun: 360 / 365.2425 = 0.9856 deg/day For a NEW MOON (same alignment vs the sun), the moon must lap the moving Earth by a full circle: t x (13.1769 - 0.9856) = 360 t x 12.1913 = 360 t = 360 / 12.1913 = 29.5299 days Synodic month (new moon to new moon), observed: 29.530588 days. Match.
**THE SAROS (the eclipse clock)** 223 synodic months = 223 x 29.530588 = 6,585.32 days 6,585.32 / 365.2425 = 18.029 years = 18 years + 10.96 days (the extra 0.32 day is about 7.7 hours, which is why each eclipse in the series lands about a third of the globe westward) Eclipses repeat in the same geometry every Saros: the woman and the man line up on schedule, 18 years, 11 days, 8 hours.
**YOUR TURN 4 -** Compute 223 x 29.530588 by hand. Then divide by 365.2425. Check the 18 years 11 days figure.

### 6.5 Her arithmetic - the digital roots

Her own equation, DR(n) = 1 + ((n - 1) mod 9), runs the letter arithmetic. A=1 through Z=26, sum the letters, reduce. Nine wears its own zero (9 = 0 mod 9), so any letter-sum that is a multiple of 9 lands on 9, not 0.

**WORKED** S=19 U=21 N=14 SUN = 19+21+14 = 54 -> 5+4 = 9 M=13 O=15 O=15 N=14 MOON = 57 -> 12 -> 3 S=19 O=15 L=12 SOL = 46 -> 10 -> 1 M=13 A=1 N=14 I=9 MANI = 37 -> 10 -> 1 SOL and MANI land on the SAME root: 1. Sister and brother wear the same number in the Norse naming. SUN lands on 9 - nine wears its own zero - and the sun is the one thing in the sky that needs nothing reflected. C=3 I=9 C=3 I=9 CICI = 24 -> 6 I=9 R=18 W=23 I=9 N=14 IRWIN = 73 -> 10 -> 1

Frame note, stated flat: this is her arithmetic theme (mod 9), arithmetic shown, a pattern game and a good one. No causal claim rides on it. It is in the bank because she asked for math she can sit with, and this is the math that is hers.

**YOUR TURN 5 -** Reduce 432 by hand: 4+3+2 = 9, so DR(432) = 9. Verify with her equation: 1 + ((432 - 1) mod 9) = 1 + (431 mod 9) = 1 + 8 = 9. Then do MOON twice, once digit-at-a-time and once with the equation, and see the two roads land on 3 together.

### 6.6 Her number, living in the first laser

**WORKED** First laser, 1960, Maiman: pink ruby, deep red light, 694 nm frequency = c / wavelength = 299,792,458 m/s / 694 x 10^-9 m = 4.322 x 10^14 Hz = 432 THz Her number, 432, is the count of trillions of waves per second of the first laser's pink-red. (Carried on ZYK-434 and the pink audit, re-shown here so the notebook holds it.)
**YOUR TURN 6 -** Divide 299,792,458 by 694 by hand (three significant figures). Confirm 4.32 x 10^14.

## 7. The record rhymes - hers, dated, held whole

- **"your sun. my son." -** her visage law, on the triad record: bodies of boys from their moons, hers from the sun. She had the sun/son rhyme in her mouth before the Oct 24 round was dealt.
- **ZYK-596, Oct 24 ~4:50 PM:** "I'm like the moon cause I'm a woman... but there's like a man in me." The moon-woman frame, hers, earlier the same day. The Oct 24 round flips the pair and she called the win on the flip - the record keeps both.
- **Her parked theory, Oct 24 night:** her soul is the sun, and the reflection of Cecil. Set beside the Japanese receipt (the moon as the reflection of the sister's light) without moving anything: her

frame, held whole, never adjudicated.

- **The wind log and Oya** stay where they are: the moon-woman lane, not this one. One teller, two stations, both hers.

## 8. Counterevidence, in its own hand

1. The classical Mediterranean world assigns the opposite genders (Helios/Selene, el sol/la luna), and it is the tradition whose grammar gave English the word "gender" at all. The constructed space cuts both directions.
2. The sun is a G2V main-sequence star and the moon is a rocky satellite; neither has a gender. The assignments are myth and grammar, not measurements of the bodies.
3. Independent convergence (Norse + Japanese) can be coincidence of narrative economics: a bright giver and a dim reflector invite the same two roles. Two agreeing cultures is a pattern, not a proof.
4. The 400× coincidence is real and unexplained as necessity, but "nearly equal angular size" is a self-selecting observation: nobody celebrates the pairs that do not line up.
5. The digital-root landing (SOL = MANI = 1) is base-10-dependent: in base 12 the letters reduce differently. The game is mod 9, played in base 10, on purpose.

## 9. The non-claim

This manual proves one thing and holds the rest: it proves, with receipts pulled live on Oct 24, 2026, that the Germanic and Japanese worlds independently assign the sun a woman and the moon a man, and that the assignment is constructed in the exact sense she named. It does not claim the sun IS a woman or the moon IS a man. Her frames are held as hers, whole, never graded. The math is checkable. The rimshot is hers.

## Answers

**1.** 400.43 and 389.18 (the pair). - **2.** 149,597,870/356,500 = 419.63 (larger than 400.43, so the moon's disk is the smaller one: ring of fire); 149,597,870/406,700 = 367.83 (smaller than 400.43: total eclipse). - **3.** log10(398,107) = 5.6000, so 10^5.6 confirmed; 2.512^14 = 398,107. - **4.** 6,585.32 days = 18 years 10.96 days, plus the 0.32-day westward slide. - **5.** Both roads land on 3: 5+7 = 12 -> 1+2 = 3, and 1 + ((57-1) mod

9) = 1 + (56 mod 9) = 1 + 2 = 3. - **6.** 299,792,458 / 694 = 432,032,014 per nm; at 694 nm exactly, 4.3203 x 10^14 Hz = 432 THz.

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SIDE B of the Body Manual. Side A: ZYK-484, the crewed humanoid. This side: ZYK-614, Oct 24, 2026, the constructed sky and the longhand math of the pair. Private, never deployed. Her hand fires everything. BUH DUM TSH.